

Hand Detection

Importing Libraries

import cv2

import mediapipe as mp

Used to convert protobuf message to a dictionary.

from google.protobuf.json_format import MessageToDict

Initializing the Model

mpHands = mp.solutions.hands

hands = mpHands.Hands(
 static_image_mode=False,
 model_complexity=1,
 min_detection_confidence=0.75,
 min_tracking_confidence=0.75,
 max_num_hands=2)

Start capturing video from webcam

cap = cv2.VideoCapture(0)

while True:

Read video frame by frame

success, img = cap.read()

Flip the image(frame)

```
img = cv2.flip(img, 1)
```

```
# Convert BGR image to RGB image
```

```
imgRGB = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
# Process the RGB image
```

```
results = hands.process(imgRGB)
```

```
# If hands are present in image(frame)
```

```
if results.multi_hand_landmarks:
```

```
    # Both Hands are present in image(frame)
```

```
    if len(results.multi_handedness) == 2:
```

```
        # Display 'Both Hands' on the image
```

```
        cv2.putText(img, 'Both Hands', (250, 50),
```

```
                    cv2.FONT_HERSHEY_COMPLEX,
```

```
                    0.9, (0, 255, 0), 2)
```

```
    # If any hand present
```

```
    else:
```

```
        for i in results.multi_handedness:
```

```
            # Return whether it is Right or Left Hand
```

```
            label = MessageToDict(i)
```

```
            ['classification'][0]['label']
```

```
if label == 'Left':
```

```
    # Display 'Left Hand' on
```

```
    # left side of window
```

```
    cv2.putText(img, label+' Hand',
```

```
                (20, 50),
```

```
                cv2.FONT_HERSHEY_COMPLEX,
```

```
                0.9, (0, 255, 0), 2)
```

```
if label == 'Right':
```

```
    # Display 'Left Hand'
```

```
    # on left side of window
```

```
    cv2.putText(img, label+' Hand', (460, 50),
```

```
                cv2.FONT_HERSHEY_COMPLEX,
```

```
                0.9, (0, 255, 0), 2)
```

```
# Display Video and when 'q'
```

```
# is entered, destroy the window
```

```
cv2.imshow('Image', img)
```

```
if cv2.waitKey(1) & 0xff == ord('q'):
```

```
    break
```